

ADDENDUM — Combined Feature Enhancements (Addendum 1 + Addendum 2)

Dhanashree Crop Solutions — Field Force Mobile Application

Extends: DhanashreeMasterPlans.docx (Module 1 – Crop Master, Module 2 – Farmer Master, Module 3 – Sales Person, Module 4 – Regional Manager & Back-end) | Reference: Product Recommendation & Feature Specification

This file combines all feature enhancements requested after the original Product Recommendation Document, in two batches: Addendum 1 (Sections 10.1–10.5) and Addendum 2 (Sections 10.6–10.8). Numbering continues the master document's structure so this can be included directly in the schema reference alongside the four master plans (Crop Master, Farmer Master, Sales Person Master, Regional Manager & Back-end Plan).

Addendum 1 — Route Planning, Big Farmer Directory, Auto Pop-Up & Content Admin

The following four enhancements were requested after the initial recommendation and are appended as new sub-modules 10.1–10.5. They extend Module 2 (Farmer Master), Module 3 (Sales Person), and Module 4 (Regional Manager & Back-end Team), and introduce a new content-governance role.

10.1 Farmer-wise Route Plan (Location-to-Location Journey Planning)

Purpose: When a field staff member travels from one location to another (e.g. Nashik to Sinnar), the app should plan the route and surface every farmer, plot, crop, and growth stage that falls along or near that route — turning a routine commute into a productive visit itinerary.

10.1.1 Route Planning Fields

Priority	Field	Details
Required	Start Location	Auto-detected GPS or manually entered (e.g. Nashik)
Required	End Location	Manually entered or selected from territory list (e.g. Sinnar)
System	Route Path	Calculated via Google Maps Directions API; supports multi-stop routes
System	Corridor Radius	Configurable buffer distance (e.g. 2–5 km) either side of the route within which farmer plots are matched
System	Farmers-on-Route List	All farmers/plots whose geotagged GPS falls within the corridor, sorted by distance along the route
Required	Crop Filter	Restrict the route list to a chosen crop and/or crop growth stage (e.g. show only grape plots between Nashik and Sinnar, or grape crop that is in the flowering stage now)
System	Crop-Stage Summary	For each farmer on the route: current crop, current growth stage, and plot acreage
System	Route Acreage Summary	Aggregated acreage under each crop along the selected corridor (e.g. "142 acres of grapes, 38 acres of onion between Nashik and Sinnar")
System	Highlight Farmers by Overdue Days	Mark the farmers as Green (visit not overdue), Orange (overdue by 7 days), or Red (overdue by 8+ days)

Priority	Field	Details
Action	Add Stop from Route	Staff taps a farmer on the route list to add a planned visit stop
Action	Sequence Stops	Reorder visit stops along the route for an efficient loop
Action	Start Navigation	Launch turn-by-turn navigation to the next stop

Recommendation: Build this as a map-first screen — draw the route line from start to end, overlay farmer-plot pins colour-coded by overdue days, and list plots in a scrollable panel below the map showing crop, stage, and acreage. Let staff filter the corridor by crop and crop stage before planning stops, and let the corridor radius be configurable per territory since road density varies. Offline note: route calculation requires connectivity, but once a route plan is generated it should cache the farmer list, plot locations, and crop/stage data locally so staff can view and log visits offline while travelling.

10.2 “Top / Big Farmer” Identification in a Village (Acreage-Ranked Directory)

Purpose: When a field staff member is visiting or planning to visit a village, they should be able to instantly see which farmers in that village hold the largest cultivated acreage — the highest-value relationships to prioritise — with filters to narrow the list.

10.2.1 Big Farmer Directory Fields

Priority	Field	Details
System	Total Acreage per Farmer	Sum of all plot areas linked to a farmer, auto-computed from Plot records (uses the geotagged/entered acreage values from 2.6)
System	Village Ranking	Farmers within the selected village sorted descending by total acreage
Required	Filter — Crop	Show ranking for a specific crop only (e.g. biggest grape growers in the village)
Required	Filter — Acreage Range	e.g. show only farmers above 5 acres, or between 2–10 acres
Required	Filter — Growth Stage	Show only farmers whose crop is currently at a chosen stage (e.g. flowering)
Required	Filter — Location	Village / Taluka / District selector to scope the ranked list
System	Farmer Quick Card	Name, mobile, total acreage, primary crop(s), current stage, last visit date — tappable to open full profile

Recommendation: Surface this as a “Village Insights” tab: staff selects (or is auto-located into) a village, and the app shows a ranked, filterable list — biggest farmer first. Combine all four filters (crop, acreage, stage, location) so staff can, for example, find “grape farmers in Sinnar village, above 3 acres, currently in flowering stage” in a few taps. This directly supports prioritising high-acreage farmers for visits, recommendations, and promotional pushes.

10.3 Location-Aware Automatic Promotion Pop-Up

Purpose: Extend the existing Digital Promotion Library and content-suggestion engine (see 2.3 and 2.11) so that relevant content surfaces automatically and proactively the moment a field staff member is physically at or opens a matching farmer/plot — not just when they manually search the library.

10.3.1 Auto-Pop-Up Behaviour

Priority	Field	Details
System	Trigger — Farmer Profile Open	Opening a farmer record automatically matches the farmer's current crop + growth stage against the Promotion Library and shows matching content
System	Trigger — Geo Check-In	When GPS detects the staff member has arrived at a geotagged plot (e.g. a grape plot at flowering stage), the app auto-surfaces the matching promotion as a pop-up card
Example	Illustration	Staff visits a grape field currently at flowering stage → app automatically pops up the grape + flowering-stage video/image/WhatsApp template, ready to send with one tap
Required	Match Logic	Crop + Growth Stage is the primary match key; weather-related advisory is also a potential criteria; Language and Region can further refine which content variant is shown
Action	One-Tap Send from Pop-Up	Staff can send the surfaced content directly to that farmer via WhatsApp/SMS without leaving the pop-up
Action	Dismiss / Snooze	Staff can dismiss the pop-up or snooze it if not relevant right now
System	Logged Automatically	Every auto-suggested send is logged to the farmer's Activity & Communication Log (see 2.8), identical to a manually pushed promotion

Recommendation: This is a proactive layer on top of the existing suggestion engine (2.3/2.11) — the current design shows suggestions when staff opens a farmer; this addendum makes it also trigger on geo-arrival at the plot, so staff are prompted even before they manually open the record, useful when walking a village and moving farmer to farmer.

10.4 Content Admin Role & Governed Promotion Push

Purpose: Introduce a dedicated Content Admin role that centrally controls what promotional content reaches the sales team and, in turn, what the sales team can forward to farmers — with full traceability and a built-in frequency cap to avoid over-messaging farmers.

10.4.1 Content Admin — Role & Permissions

Priority	Field	Details
Required	Role: Content Admin	New role in the RBAC model (alongside Field Staff, Regional Manager, Back-end Admin — see 4.2)
Required	Exclusive Send Authority	Only the Content Admin can push digital media/content out to the sales team; field staff cannot originate new content, only forward what has been approved to them

Priority	Field	Details
Required	Targeting Criteria	Content Admin selects target audience by Crop, Growth Stage, Region (territory/village), and Weather-Related Event (e.g. an incoming frost or heavy-rain advisory)
System	Push to Sales Team	Content is pushed to the relevant field staff (filtered by the targeting criteria above), who then relay it to their matching farmers
System	Direct-to-Farmer Send	Content Admin can alternatively create rules to send content directly to farmers for a specific crop + stage + weather event, bypassing field staff
System	Auto-Logged in Activity Timeline	Any content the Content Admin sends for a specific crop/stage is automatically recorded in the affected farmer's Activity & Communication Log (see 2.8), labelled "Content Team" so it's distinguishable from field-staff-sent messages

10.4.2 Per-Farmer Messaging Frequency Cap

Priority	Field	Details
Required	Cap Rule	Content Admin may send a maximum of 2 messages per week, per farmer per crop, from the Content Team channel. This number of messages per week is Configurable by the system admin and is not hardcoded
System	Cap Counter	App tracks Content-Team sends per farmer on a rolling 7-day basis
System	Cap Enforcement	Once a farmer has received 2 Content Team messages in the current week, further sends are blocked/queued until the window resets. The blocked/queued message is visible in the farmer's Activity Log to the relevant user, who can trigger it manually; otherwise the message stays unsent
System	Cap Warning	Content Admin sees a warning at send time listing which farmers in the selected audience have already hit the weekly cap
Required	Configurable Limit	Recommend making the "2 per week" figure an admin-configurable setting (per back-end control, see 2.15) rather than hard-coded, in case policy changes later

Recommendation: Keep the cap independent of field-staff-initiated sends (2.11) — the 2-per-week limit applies specifically to the centrally-pushed Content Team channel, so a farmer can still receive individual recommendations from their assigned field staff without those counting against the Content Team's cap. RBAC Update: Update the Role-Based Access Control table (4.2) to add Content Admin — access to the full Promotion Library, audience targeting tools (crop/stage/region/weather), and push analytics; no access to farmer visit/call logs beyond what is needed to confirm delivery.

10.5 Summary of Addendum 1 Additions

#	Addition	Extends
10.1	Farmer-wise route plan between two locations, showing crop area, stages, and acreage along the corridor, with a crop filter	Module 2 (Farmer/Plot), Module 3 (Field Staff)
10.2	Village-level “big farmer” directory ranked by acreage, filterable by crop, acreage, stage, and location	Module 2 (Farmer Master)
10.3	Automatic promotion pop-up triggered by farmer-profile open or geo check-in at a matching crop + stage plot	Module 1 (Promotion Library), Module 3 (Field Staff)
10.4	New Content Admin role with exclusive send authority, crop/stage/region/weather targeting, automatic activity-timeline logging, and a 2-messages-per-week-per-farmer cap	Module 4 (Admin Controls), Section 4.2 (RBAC)

These additions should be reflected in the database schema (new tables/fields for route corridors, acreage aggregation views, geo-trigger rules, and a content_team_sends table with a rolling-week counter) and included in the Phase 2 and Phase 3 roadmap items (Section 7), since route planning and the big-farmer directory depend on Farmer/Plot data already scoped for Phase 1, while the Content Admin role and governed push logic extend the Phase 2 promotions work.

Addendum 2 — Weather Advisory & Market Price Intelligence Enhancements

Two further enhancements were requested after Addendum 1 and are appended here as new sub-modules 10.6 and 10.7. Numbering continues the master document's structure for direct inclusion in the schema reference.

10.6 Location-Aware Weather Advisory Pop-Up (Current Visit Weather & Forecast)

Purpose: While a field staff member is at or near a farmer's geotagged plot, the app should automatically pull the current weather and short-range forecast for that exact location, match it against crop + growth-stage risk rules, and surface a ready-to-send advisory in the same pop-up mechanism already defined for crop-stage promotions (see 10.3). This turns the field visit into a real-time, location-specific advisory moment rather than a generic broadcast.

10.6.1 Weather Data Capture at Visit

Priority	Field	Details
Required	Live GPS Capture on Visit	App reads the staff member's current GPS position when a farmer/plot record is opened or a visit is logged
System	Current Weather Snapshot	Fetches from weather API for the visit's exact coordinates — temperature, humidity, rainfall, wind speed, conditions
System	Short-Range Forecast	Next 24–72 hour forecast for the same coordinates — rainfall probability, temperature range, frost/heat risk
Required	Crop + Stage Risk Rules	Linked to Crop Master (2.4) — e.g. Flowering-stage grape + rainfall forecast > 20mm → fungal-risk advisory

Priority	Field	Details
System	Advisory Match Engine	Cross-references live weather + forecast against the farmer's current crop/stage risk rules for that plot
System	Auto-Pop-Up Card	If a match is found, an advisory card appears automatically — same UI pattern as the crop-stage promotion pop-up (10.3)
Action	One-Tap Send to Farmer	Staff sends the surfaced weather advisory to that farmer via WhatsApp/SMS without leaving the pop-up
Action	Dismiss / Snooze	Staff can dismiss or snooze the advisory if not relevant at that moment
System	Logged Automatically	Every weather advisory sent is logged to the farmer's Activity & Communication Log (2.8), tagged "Weather Advisory"
Optional	Manual Weather Check	Staff can open a "Weather" tab on any farmer/plot to view current conditions and forecast on demand, even without a triggered advisory

10.6.2 How This Extends the Existing Promotion Pop-Up (10.3)

The weather advisory pop-up is not a separate feature from the auto-pop-up already specified in 10.3 — it is a second trigger source feeding the same pop-up card component:

- Trigger A (existing, 10.3): Farmer-profile open, or geo check-in at a matching crop + stage plot → surfaces Digital Promotion Library content.
- Trigger B (new, 10.6): Geo check-in + live weather/forecast at that plot matches a crop + stage risk rule → surfaces a weather advisory instead of, or alongside, the promotion content.
- Both triggers can fire together — e.g. a grape/flowering-stage plot may show the scheduled promotional video and a rain-risk advisory in the same session, stacked as two cards.

10.6.3 Weather-Triggered Bulk Push (Territory-Wide)

Priority	Field	Details
System	Territory Weather Watch	Back-end/Content Admin monitors forecast feeds across all territories, not just individual visits
Action	Bulk Advisory Push	When a severe-weather event is forecast for a region (e.g. frost warning across Sinnar taluka), Content Admin (10.4) can push the advisory to all matching crop+stage farmers in that region in one action
System	Auto-Trigger Rule	Optionally fully automated — e.g. "Rainfall forecast > 50mm in next 24h → auto-push frost/fungal advisory to all flowering-stage farmers in the affected villages" (extends Crop Master 2.4 weather trigger rules and Phase-2 roadmap item)
System	Cap-Aware	Bulk weather pushes count against the same Content-Team frequency cap defined in 10.4.2, so farmers are not over-messaged even during an active weather event

Recommendation: Build this as a background service — a scheduled job checks each active plot's forecast against its crop+stage risk rules every few hours, and queues advisory pop-ups so they are ready the moment a staff member opens that farmer or arrives at the plot. Individual, at-the-plot advisories (10.6.1) are staff-triggered and personal;

territory-wide advisories (10.6.3) are Content-Admin-triggered and broadcast — both share the same underlying weather feed and risk-rule engine.

10.6.4 Weather Data Source

Layer	Recommendation
Current Conditions	OpenWeatherMap Current Weather API or IMD (India Meteorological Department) API
Short-Range Forecast	OpenWeatherMap One Call API (hourly/daily) or IMD forecast feed
Severe Weather Alerts	IMD nowcast/warning bulletins where available, as a supplementary trigger source
Refresh Frequency	On-demand at visit/check-in; background refresh every 3–6 hours for territory-wide risk monitoring

Note: This upgrades the Phase 2 "Weather Trigger Rules" item already listed under Crop Master (2.4) and the Phase-2 roadmap item "Weather-trigger bulk push rules" (Section 7) from a territory-level future feature to a per-visit, location-specific live feature — the underlying weather feed and risk-rule engine are the same; 10.6 adds the real-time, GPS-tied pop-up layer on top.

10.7 Market Price Intelligence Dashboard (Crop Rate Trends & Forecast)

Purpose: Add a dashboard module that tracks market (mandi) rates for each crop over time — from 1 Jan 2025 up to the current date — so that price movements are visible at a glance, and so the app can proactively point field staff and farmers toward crops whose prices are trending upward. This gives farmers historical context and a rate forecast to support better selling and planting decisions.

10.7.1 Market Rate Data Fields

Priority	Field	Details
Required	Linked Crop	Each price record is tagged to a Crop Master entry (2.1) — e.g. Grapes, Tomato, Onion
Required	Market / Mandi Name	Source APMC market or mandi the rate was recorded at
Required	Rate Date	Date the price point applies to
Required	Price (Min / Max / Modal)	Per-quintal or per-kg rate range and modal (most common) price, as typically published by APMC/eNAM
Optional	Variety / Grade	Where rates differ by grade (e.g. Grapes — Thompson Seedless vs Sonaka)
System	Rate History Store	All historical rate points retained from 1-Jan-2025 onward, per crop per market, to power trend charts
System	Daily/Periodic Rate Sync	Scheduled import from external market-rate source (see 10.7.4) rather than manual entry
Optional	Manual Rate Override	Back-end/Content Admin can manually add or correct a rate point if the external feed is delayed or a local market isn't covered

10.7.2 Dashboard — Price Trend & Farmer Match View

Priority	Field	Details
System	Crop Price Trend Chart	Line chart per crop showing modal price from 1-Jan-2025 to the current date, with the ability to select a date range or a specific market
System	Rising / Falling Indicator	Each crop tagged Rising / Stable / Falling based on the trend over a configurable recent window (e.g. last 15/30 days)
System	Price Forecast	Short-range projected price trend (e.g. next 1–2 weeks) based on the historical series, shown as a dotted continuation of the trend line
Action	"Growers of This Crop" Drill-Down	Tapping a rising crop (e.g. Grapes) opens the list of farmers in the staff member's portfolio (or territory, for Regional Managers) currently growing that crop
Required	Filter — Crop	Focus the dashboard on one crop's rate history and grower list at a time
Required	Filter — Market / Region	Compare rates across nearby mandis, or scope to the market relevant to the staff member's territory
Required	Filter — Growth Stage	Combine with crop-stage data (2.7) — e.g. show grape growers currently near harvest whose crop price is rising, since these farmers benefit most from timely market information
System	Compare Crops	Side-by-side trend lines for two or more crops, useful where a farmer is deciding between crop options

10.7.3 Farmer-Facing Advisory & Push

Priority	Field	Details
Action	Rate Advisory Push	Staff can push a plain-language rate summary ("Grape prices up 12% this month at Nashik APMC") to matching farmers via WhatsApp/SMS, using the same one-tap send pattern as 2.11/10.3
System	Auto-Suggested Rate Alert	When a linked crop crosses into "Rising" status, the app suggests a rate-advisory push to that crop's growers, alongside the existing crop-stage promotion suggestions (2.3/10.3)
System	Logged in Activity Log	Rate advisories sent to a farmer are recorded in that farmer's Activity & Communication Log (2.8), tagged "Market Rate Advisory"
System	Cap-Aware	Where sent via the centralised Content Team channel, rate advisories count against the same weekly cap defined in 10.4.2

10.7.4 Market Rate Data Source

Layer	Recommendation
Primary Source	Agmarknet (Government of India APMC price portal) daily commodity price feed

Layer	Recommendation
Alternate / Supplementary	eNAM (National Agriculture Market) API where the relevant mandi is onboarded
Historical Backfill	Bulk-import Agmarknet historical data from 1-Jan-2025 to present at launch, to seed the trend charts immediately
Sync Frequency	Daily automated pull; manual override available to Back-end/Content Admin (10.7.1)
Forecast Method	Start with a simple moving-average / trend-line projection on the historical series; evaluate a more advanced time-series model (e.g. ARIMA/Prophet) once enough in-app data has accumulated

Recommendation: Treat 1-Jan-2025 as the fixed start date for the historical baseline requested — backfill Agmarknet's published data for that full range at launch so every crop's trend chart is populated from day one, then keep it current with the daily sync. Pair the "Rising" indicator directly with the existing Village Insights / Big Farmer directory (10.2) so staff can filter for "grape growers, above 3 acres, in a village, where grape prices are currently rising" in one combined view — this turns a market signal directly into a prioritised, actionable visit list.

10.7.5 Access & RBAC

Role	Market Rate Dashboard Access
Field Staff	View rate trends and grower drill-down for their own assigned farmers only; can push rate advisories to their farmers
Regional Manager	View rate trends and grower drill-down aggregated across their territory; reporting on rate-advisory reach per staff
Content Admin	Configure auto-suggested rate-alert rules; send centralised rate advisories NOT subject to the frequency cap (10.4.2)
Back-end Admin	Manage market-rate data sources, manual overrides, and forecast configuration; full access across all crops and markets

This extends the Role-Based Access Control table (4.2) and the Content Admin permissions defined in 10.4.1.

10.8 Summary of Addendum 2 Additions

#	Addition	Extends
10.6	Location-aware weather advisory pop-up: current conditions + short-range forecast at the visit location, matched to crop+stage risk rules, plus territory-wide weather-triggered bulk push	Module 1 (Crop Master — weather trigger rules, 2.4), Module 3 (Field Staff pop-up, 10.3), Module 4 (Content Admin bulk push, 10.4)
10.7	Market Price Intelligence dashboard: historical crop rate trends from 1-Jan-2025 to current, rising/falling indicator, short-range price forecast, and drill-down to matching growers for rate-advisory push	Module 1 (Crop Master), Module 2 (Farmer/Plot), Module 3 (Field Staff), Module 4 (Reporting, RBAC 4.2, Content Admin 10.4)

These additions should be reflected in the database schema (new tables for `weather_snapshots`, `weather_risk_rules`, `market_rate_history`, and a `crop_price_trend` materialized view keyed by crop + market + date) and included in the roadmap: 10.6 (individual advisory pop-up) fits naturally into Phase 2 alongside the existing promotion pop-up work, since it reuses the same trigger/pop-up mechanism; 10.6's territory-wide bulk

push and the full 10.7 Market Price dashboard depend on the Content Admin role and reporting layer, so they are best scheduled alongside or after Phase 3.

Document prepared for: Dhanashree Crop Solutions | Reference: DhanashreeMasterPlans.docx (Addendum 2) | Website: www.dhanashreecropsolutions.com